



The Silicon Valley Workshop

Session 2 - Python & FastAPI

How workshop will be conducted?



- 1) Workshop will be interactive not one sided.
- 2) Workshop will be of 1 hour 30 minutes.
- 3) First we will cover theory than I will demonstrate and you are supposed to follow.
- 4) Best listener and performer may rewarded with swages if available.

What we will cover?



- 1) Why Python is slow and still used ?
- 2) What is Virtual Environment and Why we need them?
- 3) How to setup virtual Environment us UV and why?
- 4) What is API and RestFul API?
- 5) What is FastAPI and why it is famous?
- 6) Basic Fast API program.

Why Python is slow and still used ?



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- The GIL (Global Interpreter Lock)
- The "it doesn't matter" argument?

What are Virtual Environments and Why we need them?



A virtual environment is an isolated sandbox that allows you to install everything you need for a project without affecting things globally. Python contains everything you need to create and use a virtual environment, and it's easy.

What is UV and why use it?



In python we need a package manager.

What is package Manager?

[Reference](#)

What is UV and why use it?



UV is modern and ultra fast Package manager as it is build upon Rust.

How to setup virtual environment with UV?



- 1) Install uv ([link](#))
- 2) Verify uv –version
- 3) Add to the path.
- 4) Open vs code select a folder.
- 5) Initialize - uv init (Standard project structure)
- 6) Install required libraries:
 - a) uv add fastapi

What is API (Application Programming Interface)?



A set of rules and definitions that let software systems or modules communicate with each other.

Why APIs matter

APIs make it easier to:

- **Connect different systems** without tightly coupling them.
- **Reuse functionality** instead of rebuilding it from scratch.
- **Speed up development** by integrating with existing services.
- **Scale applications** by distributing responsibilities across services.

An API simplifies complexity by exposing only what another application needs to know –no more, no less. That clarity helps developers build faster, safer, and more maintainable systems.

APIs form the foundation of modern [software engineering](#) and architecture, enabling microservices, cloud-native apps, and third-party integrations across platforms.



Example:

Let's say that you want to use your mobile app to check the weather. When you click, the app sends a request to a weather service API, which receives that request, processes it, and returns the current forecast. The app then displays the data to you. This exchange happens over the internet, often within milliseconds

Type of APIs Base on:



- 1) APIs based on availability and access
- 2) APIs based on where they run (Web or remote API)
- 3) API protocols and styles:
 - a) REST (Representational State Transfer)
 - b) SOAP (Simple Object Access Protocol)

In everyday apps, APIs power:

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- **Social media platforms:** APIs let apps post to your feed, pull in friend lists, or authenticate with your account. For example, logging in with GitHub or Facebook uses an API call to confirm your identity.
 - **Travel and booking:** When a travel site shows you flights from different airlines, it's pulling real-time data from multiple APIs. Payment processing and booking confirmations are also API-driven.
 - **Weather and mapping services:** APIs provide up-to-date weather forecasts, GPS directions, and location-based search. They help mobile apps deliver timely, location-aware experiences.

In development and DevOps, APIs support:



- Cloud services
- CI/CD pipelines
- AI-assisted development

RESTFul API



It is an architecture to make apis which are more flexible and efficient in many cases than other apis like SOAP.

To make RESTFul API we have to follow some architectural principles:

- 1) Uniform Interface
- 2) Statelessness nature
- 3) Client - server decoupling
- 4) System Layered Architecture
- 5) Cacheability

What is FASTAPI?



FastAPI is modern python web framework which uses REST api architecture to build RESTFul apis.

Why FastAPI?

- 1)Speed of Development - Easy to code
- 2)Performance - Due on **Starlette and Pydantic.**
- 3)Automatic Docs - Swagger UI for testing endpoints